Development of a force myography sensor for PrHand prosthesis activation using fiber bragg grating sensor and 3D printing

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***Abstract*—The human-prosthesis interaction is an important factor for the implementation of any assistive device. Some works discusses how to use force myography (FMG) signals in the context of prosthesis, although, of particular interest is to use it in the upper-limb prosthesis PrHand. The sensor in conjunction with the reading system manages to send signals to contract and expand a linear motor as first step for prosthesis control. Aiming to improve its current state, making it easiest to manage for amputee people.**

***Index Terms*—Force myography, fiber bragg grating, 3D printing, prosthesis, upper-limb.**

I. Introduction

Around 75.000 amputations were performed in Brazil on the 2022 [1] and it is expected that at the 2050 the number of total amputee patients increases to 3.6 million in the United States of America [2]. Amputation is mainly caused by a type of trauma, medical illness, or congenital condition. This operation is life-changing since the surgeon must create a new organ contact with the environment. Depending on the severity, an amputation affects the performance of the activities of the daily living, and leaves the person with a deterioration of physical, psychological, and social functions [3], [4].

Biological signals have been studied to be employed in the control of prostheses, for EMG (Electromyography), FMG (Force Myography) and EIT (Electrical Impedance Tomography) signals allows to detect limb and nerve movements [5].

Since the control of the flexion and extension of the prosthesis is the main goal of this work, muscles used in the managment of those gestures must be monitored, such as the flexor digitorum profundus (FDP) and flexor digitorum superficialis (FDS) [6]. For that reason an FMG sensor will be developed, the overall system consist of a flexible piece with an FBG (fiber Bragg grating) embedded, connected to an interrogator where the information of the sensor can be read for its use in mechanical objects activation.

The design of the proposed sensor is based on the work presented by Cátia Tavares, *et al*. [7]. However, some changes were performed.

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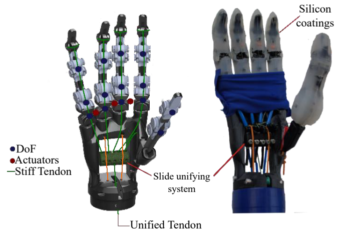


Fig. 1: PrHand prosthesis based on soft-robotics.

II. Methodology

A. Materials

The PrHand (Figure 1) is an upper-limb prosthesis based on soft robotics which aims to assist amputee patients. This prosthesis has 15 Degrees of Freedom (DoF) and was developed with two actuation systems, a servomotor to perform flexion and extension movements, and pneumatic actuators to perform the abduction and adduction movements [8].

A single FBG was inscribed in a photosensitive single mode optical fiber (GF1, Thorlabs) using the phase mask method, with a frequency of 10 Hz, energy of 0.37 mJ and centered at 265.524 nm, with 3 mm length (DPS-266-Q, Changchun New Industries). For the data collecting, an interrogator (HYPERION si255, Luna's) with measurement frequency of the 5000 Hz and wavelength range from 1500 to 1580 nm was used. The FBG was covered with a 3D-printed structure of transparent flexible filament (Flex, 3DFila). Also, cyanoacrylate glue (Super Bonder) was used to improve the FBG fixation into the flexible piece.

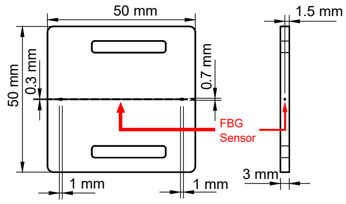
The PrHand is actuated with a linear actuator (L16-P, Actuonix), and a dual full-bridge driver (L298N, STMicroelectronics) is placed to archive the flexion and extension. A single board computer (SBC, Raspberry Pi 4) was implemented for the mechanical activation of the prosthesis.

B. Methods

The size of the enclosure is 50 x 50 x 3 mm. At a height of 1.5 mm, it has a 0.3 mm hole where the FBG is placed. In addition, it has two slots of 0.7 mm diameter and 1 mm height, centered with the hole. The design also includes two spaces parallel to the FBG to put a elastic band to secure the sensor in the muscle, illustrated in figure 2a.

For the 3D printing, the infill density was placed at 60% with a pattern of grid, all the other configuration were placed according to the manufacturer's recommendations.

To build the sensor, the first thing to do is pause the 3D-printing in certain layer (For this sensor the layer eight was used). Then, the FBG was placed in the middle of the piece and aligned with the groove. Once placed, the optical fiber must be secured to the printer plate with tape, for it further fixation in each groove with glue. At last, let it dry and restart the 3D-printing.



(a)

(b)

Fig. 2: (**a**) Sketch of the enclosure. (**b**) FMG sensor.

Once the process of printing its done as shown in Figure 2b, the FBG sensor can be tested in the interrogator. The data of every peak is stored using an API provided for the interrogator manufacturer. Once enough data is acquired for each gesture, the mean of the Bragg wavelength is calculated, and the process of identification can be done, where the difference between the actual value versus the two means of the gestures is the factor to chose if the prosthesis opens or closes.

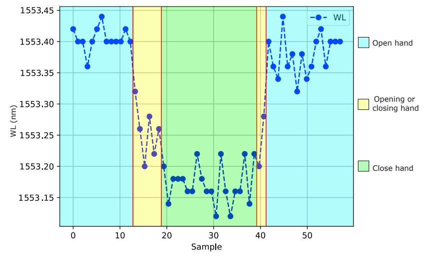
As this sensor measures the overall volumetric variation of the FDP and FDS, the individual contribution of each will not be considered.

III. Results

The test was made with a non-amputee female subject, placing the sensor in the upper part of the forearm. The sensor was attached with an elastic band at a contraction that was comfortable for the subject, as presented in Figure 3a. The variation of the Bragg wavelength for the open and closed hand state, and the intermediate states are easily observable in Figure 3b.

In the moment that any action is identified, the SBC sets the correspond voltage in the control pins of the driver to achieve the movement. Allowing to control the flexion and extension of the prosthesis using an FMG signal of the forearm.

(a)



(b)

Fig. 3: (**a**) sensor placement. (**b**) Gesture comparison by its Bragg wavelength.

IV. Discussion and Conclusions

This works shows a functional FMG sensor based in optical technologies, embedded in a 3D printed piece for mechanical object activation.

In the construction process, choosing the right stop layer is a critical and mandatory step for its correct development, this because every 3D printer has different properties and parts, knowing well this characteristics allow avoiding mistakes such as the rupture of the fiber by the extruder. The glue is also important to ensure the correct strength transfer to the fiber.

As future works, in order to increase the comparability with other upper-limb sensors, the calculation of accuracy and precision must be performed. Also, its validation with persons who have had amputations must also be realized. For the recognition of complex gestures more than a single FBG must be used, since its required the monitoring of multiple muscles. During the experiment, it was found that the sensor does not depend on the force applied to distinguish one gesture from another. However, this phenomenon will be carried out in the future.

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